Real-Time Emotion Classification

Javier Guapilla-Diaz, McKinley Nhi Seecof Quevedo

Objective:
Train a neural network model capable of recognizing human emotion from facial expressions and use it for real-time emotion classification.

Background: Being able to read human emotion is a classification task our minds do automatically. This comes in handy during the conversations or arguments we have, helping steer our thought process, and consequently, the words we say. While this is somewhat easy one-on-one, the task begins to increase in difficulty as we increase the number of people we are trying to read or simply by trying to read someone we do not know.
In our project, we aim to use a Convolutional Neural Network (CNN) to classify faces in real time, using our computer's camera, into one of 7 categories: anger, disgust, fear, happiness, sadness, surprise, and neutral. Uses for this vary, from pain assessments in the medical field, driver monitoring (sleepiness), or consumer screening in market analytics.

Data/Features: FER-2013

- 28,709 (48x48 pixel grayscale) pictures of human faces.
- Pre-categorized into six different emotions (anger, disgust, fear, happiness, sadness, surprise, and neutral).
- Images taken from various angles but pre-sized so the face occupies about the same amount of space in each image.

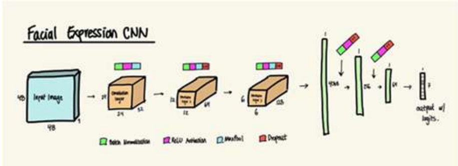
For training we
- Rescale pixel values from 0-255 to 0-1.
- Transform some images for more robust classification.

For the real-time component:
- using input data from the camera we detect the face(s), draw boxes around them, lower resolution to 48x48, and transform to grayscale

Model Architecture

CNNs are made up of three main types of layers: Convolutional layers, Pooling layers, and Fully-connected layers.

Convolutional Layer: Convolutional layers accept 3D input corresponding to an image's height, width, and depth (for color images RGB). A filter (or kernel) is then moved across the image and a dot product is calculated between the input pixels and the filter. The filter is moved across the entire image to create a feature map. More than one filter can be used to increase the depth of the output. The stride (number of pixels) that the kernel moves over the input matrix and padding can also increase or decrease the size of the output. The weights of the filter are learned during training. After each convolution operation, a Rectified Linear Unit (ReLU) transformation is applied to the feature map.
Pooling Layer: The pooling layer also moves a filter over the input but instead of performing a convolution with weights, it performs an aggregation function. Max pooling selects the pixel with maximum value to send to the output. Average pooling calculates the average value to send to the output.
Fully-connected Layer: The fully-connected layer connects each input node to each output node. This layer is used for the final classification based on the features extracted through the previous layers.



CNN FLOW:
Input (1x48x48) ->
Conv2D (1->32, 3x3): Batch Norm. ReLU Activation. MaxPool (2x2). [32x24x24] ->
Conv2D (32->64, 3x3): Batch Norm. ReLU Activation. MaxPool (2x2). Dropout (0.25). [64x12x12] ->
Conv2D (64->128, 3x3): Batch Norm. ReLU Activation. MaxPool (2x2). Dropout (0.25). [128x6x6] ->
Flattening [4608] ->
Linear (4608 -> 256): Batch Norm. ReLU Activation. Dropout (0.5). ->
Linear (256 -> 64): Batch Norm. ReLU Activation. Dropout (0.5). ->
Linear (64 -> 7):. -> **OUTPUT**

Test Classification Accuracy

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--- Accuracy of Model ---

Testing Accuracy: 65.03204%

--- Accuracy By Class ---

Accuracy of angry: 52.47%
Accuracy of disgust: 48.62%
Accuracy of fear: 41.02%
Accuracy of happy: 85.98%
Accuracy of neutral: 63.71%
Accuracy of sad: 58.06%
Accuracy of surprise: 79.12%
```

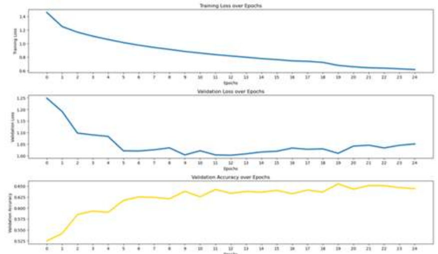
We met our goal of at least 50% accuracy. Something to note is the lower accuracies in classifying **fear** and **disgust**, possibly due to both their lesser presence in the dataset, but also their similarity. In contrast, we have **happy** and **surprise** with the greatest accuracies, which we believe is due to how different they are from other classes.

Real-Time Pipeline

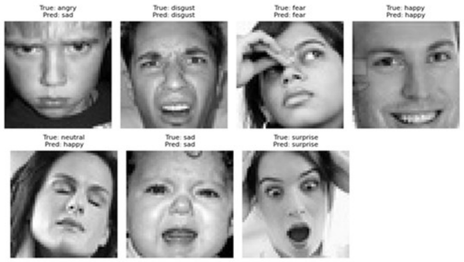
To demonstrate how our model could be used in the real world we designed a script for classifying emotion in real time using a computer's camera.

- We use a pre-trained face detection neural network model (YOLOv11n) to detect the face(s) in view of the camera and draw the bounding boxes around the face.
- We convert the image inside the bounding box to grayscale, resize it to 48x48, and scale the pixels to be between 0-1.
- We then feed that image into our trained model and display the emotion classification along with the confidence of the prediction.

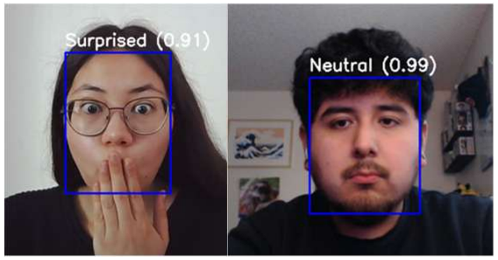
Results: Training & Validation Loss



Images: Actual VS Predicted Class



Real-time Classification



Real-time classification for expressions of surprise and neutral

Summary

In summary, we built a CNN on the FER-2013 dataset to classify 7 facial emotions in real time using YOLO for face detection. We ultimately chose a 3 block convolutional design followed by a 3 layer fully connected neural net, with a sprinkling of dropout and batch normalization to combat overfitting. After many attempts to break through our final 65% test accuracy, we settled on this design, as any attempts at altering this design resulted in worse performance.
Per class, we found that "happy" and "surprise" had the highest accuracies with about 86% and 80% accuracy on the test set respectively. Whereas "fear" and "disgust" were on the lower end of the test accuracy, being about 41% and 49% respectively. Regardless, we are happy with the results, as even the lower end beats randomly guessing a class (~14.3%) per emotion.
A limitation on our model was the data we trained on, being made up of grayscale 48x48 images, losing not only detail in resolution, but color as well. We attempted to remedy this by introducing data augmentation, however as aforementioned, we peaked at around 65% accuracy.
Something we would do differently in the future would be to use a larger dataset with more uniform representation of emotions, as well as color which would allow for better utilization of a CNN.

References:

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